

# React Machine Coding: Priority List

SDE2 / SDE3 preparation | 14 done, 39 to go (hooks grouped)

| Priority                                        | Problems | Target        |
|-------------------------------------------------|----------|---------------|
| <b>P0</b> Must do (asked most often, core SDE2) | 12       | Week 1-2      |
| <b>P1</b> High priority (frequently asked)      | 12       | Week 3-4      |
| <b>P2</b> Good to have                          | 12       | Week 5        |
| <b>P3</b> Stretch (SDE3, rarely a full round)   | 3        | Only for SDE3 |
| <b>Already done (revise and upgrade)</b>        | 14       | Ongoing       |

**SDE2:** finish P0 and P1, then revise the done list. **SDE3:** add P2 and P3, and explain tradeoffs (state shape, performance, accessibility) as you build.

## P0: Must do (asked most often, core SDE2)

Do these first. They come up in most frontend machine coding rounds, and each one tests a pattern that other problems reuse (recursion, timers, portals, keyboard handling).

| #  | Problem                     | Difficulty | Level | Must cover in the interview                                                                                        | Done                     |
|----|-----------------------------|------------|-------|--------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1  | Todo List                   | Easy       | SDE2  | Add/edit/delete, filter (all/active/done), persist to localStorage, stable ids as keys                             | <input type="checkbox"/> |
| 2  | Countdown Timer / Stopwatch | Easy       | SDE2  | setInterval cleanup, pause/resume/reset, drift correction with Date.now(), mm:ss format                            | <input type="checkbox"/> |
| 3  | File Explorer / Folder Tree | Medium     | SDE2  | Recursive component, add/rename/delete file or folder, expand/collapse, immutable tree updates                     | <input type="checkbox"/> |
| 4  | Modal / Dialog              | Medium     | SDE2  | createPortal, close on Esc and backdrop click, focus trap, return focus, scroll lock, role=dialog                  | <input type="checkbox"/> |
| 5  | Infinite Scroll             | Medium     | SDE2  | IntersectionObserver sentinel, loading/error/end states, no duplicate requests                                     | <input type="checkbox"/> |
| 6  | Typeahead with keyboard nav | Medium     | SDE2  | Upgrade your Search: AbortController for race conditions, cache, up/down/Enter/Esc, ARIA combobox, highlight match | <input type="checkbox"/> |
| 7  | Data Table                  | Medium     | SDE2  | Sort by column (asc/desc), search, filter, pagination, useMemo for derived rows                                    | <input type="checkbox"/> |
| 8  | Toast / Notification System | Medium     | SDE2  | Context + useToast() API, queue, auto-dismiss with pause on hover, positions, max visible                          | <input type="checkbox"/> |
| 9  | Tic-Tac-Toe (N x N)         | Medium     | SDE2  | Board size as prop, O(N) winner check per move, draw, reset, undo via history                                      | <input type="checkbox"/> |
| 10 | Dropdown / Multi-select     | Medium     | SDE2  | Click outside to close, keyboard nav, search inside, selected chips, controlled API                                | <input type="checkbox"/> |
| 11 | Multi-step Form / Wizard    | Medium     | SDE2  | Config-driven steps, per-step validation, back/next keeps data, summary before submit                              | <input type="checkbox"/> |
| 12 | useDebounce + useFetch      | Hook       | SDE2  | Debounced value hook; fetch hook with loading/error/data, AbortController on unmount, simple cache                 | <input type="checkbox"/> |

## P1: High priority (frequently asked)

Next in line. These appear often at product companies and cover drag and drop, games, and state that has to stay in sync.

| #  | Problem                   | Difficulty | Level | Must cover in the interview                                                                      | Done                     |
|----|---------------------------|------------|-------|--------------------------------------------------------------------------------------------------|--------------------------|
| 13 | Grid Lights               | Medium     | SDE2  | Track click order, deactivate in reverse with timers once all are on, block clicks while running | <input type="checkbox"/> |
| 14 | Memory / Match-pairs game | Medium     | SDE2  | Shuffle, flip two, match check, lock board while comparing, moves counter, win state             | <input type="checkbox"/> |
| 15 | Tooltip / Popover         | Medium     | SDE2  | Position with getBoundingClientRect, flip at viewport edge, show delay, aria-describedby         | <input type="checkbox"/> |
| 16 | Transfer List             | Medium     | SDE2  | Select items, move left/right, select all, keep state per list                                   | <input type="checkbox"/> |

| #  | Problem                         | Difficulty | Level  | Must cover in the interview                                                                   | Done                     |
|----|---------------------------------|------------|--------|-----------------------------------------------------------------------------------------------|--------------------------|
| 17 | Shopping Cart                   | Medium     | SDE2   | useReducer, add/remove/quantity, derived totals, Context for global cart                      | <input type="checkbox"/> |
| 18 | Kanban Board                    | Hard       | SDE2/3 | HTML5 drag and drop between columns, reorder inside column, add card, persist                 | <input type="checkbox"/> |
| 19 | Drag-and-drop sortable list     | Hard       | SDE2/3 | dragstart/dragover/drop, reorder array immutably, visual drop indicator                       | <input type="checkbox"/> |
| 20 | Virtualized List (10k+ rows)    | Hard       | SDE3   | Render only visible rows: scrollTop / rowHeight, overscan buffer, fixed then variable heights | <input type="checkbox"/> |
| 21 | Calendar / Date Picker          | Hard       | SDE2/3 | Build month grid, prev/next month, min/max dates, range selection, keyboard nav               | <input type="checkbox"/> |
| 22 | Counter with undo/redo          | Easy       | SDE2   | History stack with past/present/future, limit history size                                    | <input type="checkbox"/> |
| 23 | Like Button (optimistic update) | Easy       | SDE2   | Update UI first, roll back on API failure, disable while pending, loading state               | <input type="checkbox"/> |
| 24 | Custom hooks set                | Hook       | SDE2   | useThrottle, useLocalStorage, useClickOutside, useIntersectionObserver                        | <input type="checkbox"/> |

## P2: Good to have

Quick wins and SDE3 design problems. Do the easy ones in one sitting; spend real time on useForm and headless components if you are targeting SDE3.

| #  | Problem                      | Difficulty | Level  | Must cover in the interview                                                                         | Done                     |
|----|------------------------------|------------|--------|-----------------------------------------------------------------------------------------------------|--------------------------|
| 25 | Reusable useForm hook        | Hard       | SDE3   | Register fields, validation schema, touched/dirty/errors, submit handling, minimal re-renders       | <input type="checkbox"/> |
| 26 | Headless compound components | Hard       | SDE3   | Tabs/Select/Modal via Context + compound API (<Tabs.List>), controlled and uncontrolled modes       | <input type="checkbox"/> |
| 27 | Image Gallery + Lightbox     | Hard       | SDE2/3 | loading="lazy" or IntersectionObserver, lightbox with arrows and Esc, preload neighbours            | <input type="checkbox"/> |
| 28 | Calendar day view (overlaps) | Hard       | SDE3   | Place events by time, split width for overlapping events (column packing algorithm)                 | <input type="checkbox"/> |
| 29 | Stepper / Progress Steps     | Medium     | SDE2   | Config-driven steps, active/completed states, progress line width calculation                       | <input type="checkbox"/> |
| 30 | User List from API           | Medium     | SDE2   | Loading, error, empty and retry states; shimmer; search on the fetched data                         | <input type="checkbox"/> |
| 31 | Poll / Voting widget         | Medium     | SDE2   | Vote once, percentage bars, optimistic update                                                       | <input type="checkbox"/> |
| 32 | Snake game                   | Hard       | SDE2/3 | Game loop with setInterval, direction via keydown, grow on food, collision, useRef for latest state | <input type="checkbox"/> |
| 33 | Password Strength Checker    | Easy       | SDE2   | Rules list with live ticks, strength meter, show/hide password                                      | <input type="checkbox"/> |
| 34 | Theme Switcher               | Easy       | SDE2   | Context, persist choice, respect prefers-color-scheme, no flash on load                             | <input type="checkbox"/> |
| 35 | Toggle / Switch              | Easy       | SDE2   | Controlled/uncontrolled, role=switch, aria-checked, keyboard (Space)                                | <input type="checkbox"/> |
| 36 | Small hooks                  | Hook       | SDE2   | usePrevious, useWindowSize                                                                          | <input type="checkbox"/> |

### P3: Stretch (SDE3, rarely a full round)

Asked mostly at senior levels or as discussion. Knowing the approach is often enough.

| #  | Problem                  | Difficulty | Level | Must cover in the interview                                                                     | Done                     |
|----|--------------------------|------------|-------|-------------------------------------------------------------------------------------------------|--------------------------|
| 37 | Spreadsheet / Excel grid | Hard       | SDE3  | Cell state map, formula parsing (=A1+B2), dependency graph, recompute, circular reference check | <input type="checkbox"/> |
| 38 | Rich Text Editor (basic) | Hard       | SDE3  | contentEditable, bold/italic/lists, selection and range APIs, sanitising pasted HTML            | <input type="checkbox"/> |
| 39 | Video Player             | Hard       | SDE3  | Custom play/pause/seek/volume/speed, keyboard shortcuts, fullscreen API, buffered bar           | <input type="checkbox"/> |

## Already done: revise and add follow-ups

Interviewers usually ask a follow-up after the basic version works. Add these to your existing solutions.

| #  | Problem                      | Follow-ups to add                                                                          |
|----|------------------------------|--------------------------------------------------------------------------------------------|
| 1  | <b>Accordion</b>             | Allow single vs. multiple open via prop, aria-expanded, keyboard support                   |
| 2  | <b>Star Rating</b>           | Half stars, hover preview, read-only mode, keyboard (arrow keys)                           |
| 3  | <b>Progress Bar</b>          | Use transform instead of width for animation, role=progressbar, aria-valuenow              |
| 4  | <b>Tab Form</b>              | Validate before switching tabs, show errors per tab, keep data across tabs                 |
| 5  | <b>Chip Input</b>            | Enter/comma to add, Backspace removes last, no duplicates, paste several values            |
| 6  | <b>OTP Input</b>             | Paste full code, Backspace moves back, arrow keys, digits only, auto-submit                |
| 7  | <b>Image Slider</b>          | Autoplay with pause on hover, infinite loop, dots, clean up interval                       |
| 8  | <b>Pagination</b>            | Server-side paging, page-number window with ellipsis, page size selector                   |
| 9  | <b>Search / Autocomplete</b> | Fix race condition (AbortController), move cache to useRef, key by value, add keyboard nav |
| 10 | <b>Nested Checkbox</b>       | Add indeterminate state (ref.indeterminate = true), keep parents in sync                   |
| 11 | <b>Nested Comments</b>       | Edit/delete, collapse thread, depth limit, immutable recursive updates                     |
| 12 | <b>Live Chat</b>             | Auto-scroll only when at bottom, cap message count, simulate polling/WebSocket             |
| 13 | <b>Cricket Score</b>         | Polling with cleanup, show stale-data state                                                |
| 14 | <b>Traffic Light</b>         | Config as prop, pause/resume, pedestrian button, blinking yellow night mode                |

## Checklist for every problem

- Clarify requirements first: inputs, edge cases, what "done" means.
- Plan the component tree and state shape before writing code. Keep state minimal and derive the rest.
- Get a working version in about 60% of the time, then add polish.
- Clean up timers, listeners and requests in useEffect return functions.
- Use stable ids as keys, never the array index for lists that change.
- Handle loading, error and empty states for anything that fetches data.
- Accessibility: semantic HTML, keyboard support, ARIA roles where needed.
- Make it reusable: pass config and callbacks as props instead of hardcoding.
- Talk through tradeoffs out loud (useMemo cost, Context re-renders, where state lives).